FREEDOM INTERNATIONAL SCHOOL

WORKSHEET

PHYSICS

CLASS XI

SYSTEM OF PARTICLES AND ROTATIONAL MOTION

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1. Two identical particles move towards each other with velocity 2v and v respectively. What is the velocity of the centre of mass?
2. A child sits stationary at one end of a long trolley moving uniformly with speed v on a smooth horizontal floor. If the child gets up and runs about on the trolley in any manner, then what is the effect of the speed of the centre of mass of the (trolley + child) system?
3. Show that the centre of mass of a uniform rod of mass M and length L lies at the middle point of the rod.
4. From a uniform disc of radius R, a circular hole of radius R/2 is cut out. The centre of the hole is at R/2 from the centre of the original disc. Locate the centre of gravity of the resulting flat body.
5. The moments of inertia of two rotating bodies A and B are I_A and I_B ($I_A > I_B$) and their angular momenta are equal. Which one has a greater kinetic energy?
6. If earth were to shrink suddenly, what would happen to the length of the day?
7. Why is a force is applied at right angles to a heavy door at the outer edge while closing or opening it?
8. A solid cylinder of mass 20 kg rotates about its axis with angular speed 100 rad/s. The radius of the cylinder is 0.25m. What is the kinetic energy associated with the rotation of the cylinder? What is the magnitude of angular momentum of the cylinder about its axis?
9. (i) A child stands at the centre of turntable with his two arms out stretched. The turntable is set rotating with an angular speed of 40 rpm. How much is the angular speed of the child if he folds his hands back and thereby reduces his moment of inertia to 2/3 times the initial value? Assume that the turntable rotates without friction
   (ii) Show that the child's new kinetic energy of rotation is more than the initial kinetic energy of rotation.
10. A car weighs 1800 kg. The distance between its front and back axles is 1.8 m. Its centre of gravity is 1.05 m behind the front axle. Determine the force exerted by the level ground on each front wheel and each back wheel.
11. A rope of negligible mass is wound round a hollow cylinder of mass 3 kg and radius 40 cm. What is the angular acceleration of the cylinder if the rope is pulled with a force of 30 N? What is the linear acceleration of the rope? Assume that there is no slipping.